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Predicting Protected Area Designations and Assessing Transition Risk under 30×30

Master Thesis

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20-913-166

April 2026

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Abstract

The Kunming-Montreal Global Biodiversity Framework commits nearly 200 countries to protect 30% of terrestrial land by 2030 — up from 17.6% today. Where the additional 12 percentage points will be designated is unknown in advance yet materially consequential: protected-area creation imposes land-use restrictions and can strand agricultural, mining, and infrastructure assets. This thesis develops a machine-learning framework to predict the probability that any 1 km^2 pixel becomes newly designated as a protected area within a five-year horizon, and applies it to South America, the United States, and Southeast Asia. Using gradient-boosted trees (Light Gradient-Boosting Machine; LightGBM) and Random Forests trained on a pixel–year panel covering 2000–2024, the framework achieves strong out-of-sample discrimination in South America (ROC-AUC 0.897; Precision–Recall AUC 0.645): the top 1% of predicted pixels capture 90.4% of all observed protected-area establishments during the held-out test period. Calibrated predicted probabilities serve as interpretable transition-risk scores. Forward inference on 2024 covariates reveals that under business-as-usual designation rates, South America would reach the 30% target approximately 67 years late, and the United States and Southeast Asia more than 170 years late. Cross-continental transfer experiments show that model patterns are region-specific and do not generalise across governance contexts, implying that the 30×30 expansion pathway is shaped by local political economies rather than universal ecological gradients. The results provide a spatially explicit risk tool for governments, investors, and landowners anticipating the next wave of protected-area expansion.

Keywords: protected areas; 30×30; transition risk; machine learning; LightGBM; spatial prediction; land-use change; biodiversity policy

Contents

List of Abbreviations

Abbreviation	Definition
BAU	Business-as-usual
CBD	Convention on Biological Diversity
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DPI	Database of Political Institutions
ECE	Expected Calibration Error
GBM	Gradient-Boosting Machine
GSN	Global Safety Net
HNTL	Harmonised Night-Time Lights
LightGBM	Light Gradient-Boosting Machine
LOBO	Leave-One-Biome-Out
MCE	Maximum Calibration Error
MODIS	Moderate Resolution Imaging Spectroradiometer
NDVI	Normalised Difference Vegetation Index
OECM	Other Effective Area-Based Conservation Measure
PA	Protected Area
PR-AUC	Precision–Recall Area Under the Curve
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SHAP	SHapley Additive exPlanations
SRTM	Shuttle Radar Topography Mission
V-Dem	Varieties of Democracy
WDPA	World Database on Protected Areas
WGI	Worldwide Governance Indicators
WRI	World Resources Institute

1 Introduction

The global expansion of protected areas (PAs) is a central pillar of contemporary biodiversity policy. The Kunming-Montreal Global Biodiversity Framework, adopted at COP15 in December 2022, commits nearly 200 signatory countries to designating 30% of the world's land and oceans as protected by 2030 — the so-called 30×30 target (Convention on Biological Diversity, 2022). At the time of writing, terrestrial protection stands at 17.6% (UNEP-WCMC and IUCN, 2024a), meaning governments must almost double the protected land area in under eight years. While this commitment is widely regarded as necessary to halt biodiversity loss, the rapid and large-scale PA expansion it demands carries significant economic and regulatory consequences. PA designation typically introduces binding land-use restrictions that constrain agriculture, resource extraction, infrastructure development, and urban expansion (Schuster et al., 2023). Uncertainty about where new PAs will be established therefore generates material *transition risk* — the risk that future conservation decisions strand assets or alter the viability of existing land uses (Govind and Webb, 2025).

The economic stakes are substantial. Protected-area policies impose real opportunity costs in the form of foregone rents, constrained resource extraction, and reduced land values (Reynaert et al., 2024). These costs are spatially heterogeneous: they fall disproportionately on landowners, farmers, and investors whose assets sit in locations most likely to be designated (Robinson et al., 2024). From a risk-management perspective, the key question is therefore not only how much land will be protected, but precisely where. Anticipating the spatial allocation of new PAs would allow businesses, investors, and policymakers to adjust strategies and regulations proactively, reducing the probability of sudden and costly disruptions. Nature-related financial disclosure frameworks explicitly identify PA creation as a transition risk category affecting business operations, supply chains, and land-secured lending (Govind and Webb, 2025).

Critically, historical PA designation is not spatially random. A robust empirical literature documents that governments systematically protect land that is remote, ecologically important, and economically cheap to designate — areas characterised by high elevation, steep slopes, low population density, and distance from roads (Joppa and Pfaff, 2009; Venter et al., 2018; Baldi et al., 2017; Mouillot et al., 2024). This bias reflects the political economy of conservation: designation is easiest where opportunity costs are lowest. If past patterns predict future patterns, then geographic covariates can be used to build a forward-looking transition-risk model. Recent work confirms this principle (Mouillot et al., 2024; Greenhill et al., 2024), but existing approaches remain limited: they are typically designed to map ecological suitability rather than to generate calibrated probability forecasts for government designation behaviour, they focus on single regions, and they do not produce the probability estimates required for quantitative risk assessment.

This thesis addresses these gaps by developing a multi-region machine-learning framework that explicitly targets the prediction of future PA establishment. The framework treats PA designation as a rare binary transition in a pixel–year panel, trains LightGBM and Random Forest models on historical designation patterns, calibrates the raw scores to interpretable probabilities, and deploys the calibrated model to generate forward-looking transition-risk maps for the 2025–2030 period. Three regions are studied: South America (the primary model), the United States (robustness check 1), and Southeast Asia (robustness check 2). The analysis produces three concrete outputs: (i) calibrated spatial risk scores that quantify each pixel's probability of designation within five years; (ii) hotspot maps and on-track assessments under business-as-usual, moderate, and 30×30 scenarios; and (iii) estimates of the economic exposure — measured by land use type — in the highest-risk zones.

The remainder of the thesis is organised as follows. Section 2 reviews the four strands of literature this work connects. Section 3 describes the data sources, study areas, and panel construction. Section 4 presents the modelling framework and evaluation strategy. Section 5 presents the results for all three regions. Section 6 discusses the implications for conservation policy, investor risk assessment, and the limitations of the approach. Section 7 concludes.

2 Literature Review

2.1 Biodiversity Loss and the Protected-Area Policy Response

Global biodiversity is declining at a rate unprecedented in human history. The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) Global Assessment documents that human activities have already substantially altered three-quarters of the Earth's land surface and that one million plant and animal species face extinction (Díaz et al., 2019). The interlocking drivers of this decline — land-use change, overexploitation, climate change, invasive species, and pollution — compound ecological pressure and degrade the capacity of ecosystems to sustain human well-being. The loss is not merely ecological: biodiversity underpins food security, freshwater provision, disease regulation, and climate stabilisation, making its erosion a risk to human societies as much as to nature itself.

Area-based conservation — the legal designation of land as protected from extractive and destructive uses — is the primary policy instrument deployed in response. Watson et al. (2014) synthesise evidence from across the global PA network and show that, where adequately managed, protected areas measurably reduce habitat loss and maintain higher biodiversity than surrounding unprotected landscapes. However, the same review identifies a persistent gap between the formal extent of protection and its functional effectiveness: many PAs are underfunded, cover ecologically marginal land, or lack the management capacity to deter illegal activities. The world's most threatened species and ecosystems frequently remain outside the protected network (Williams et al., 2022; Chaplin-Kramer et al., 2023).

These shortcomings provided the impetus for the Kunming-Montreal Global Biodiversity Framework, adopted at COP15 in December 2022, which commits nearly 200 countries to protecting 30% of terrestrial and marine areas by 2030 (Convention on Biological Diversity, 2022). At the time of writing, terrestrial coverage stands at 17.6% (UNEP-WCMC and IUCN, 2024a), meaning governments must designate approximately the same land area as the entire existing global PA network within eight years. The pace and spatial pattern of this expansion will largely determine whether the 30×30 commitment delivers meaningful conservation outcomes or merely extends the historical record of protecting ecologically marginal land.

2.2 Spatial Biases and the Determinants of Protected-Area Siting

A foundational empirical regularity in the conservation literature is that governments do not designate PAs at random, nor do they reliably prioritise the land most in need of protection. Joppa and Pfaff (2009) provide the canonical global analysis: across a large sample of countries, PAs are systematically concentrated at high elevations, on steep slopes, in remote locations far from roads and settlements, and on land with low agricultural productivity. These characteristics are precisely those that minimise the opportunity cost of designation, suggesting that governments protect land where protection is politically and economically easy rather than where it is ecologically most urgent.

Venter et al. (2018) extend this analysis across multiple decades and show that the bias has been remarkably persistent, with PA systems continuing to expand preferentially into economically marginal land even as global conservation commitments have strengthened. Baldi et al. (2017) characterise this pattern as "opportunity-driven" siting: designation follows the path of least resistance. Sharma et al. (2025) further demonstrate that privately managed protected areas exhibit even greater bias toward low-productivity land than public PAs, indicating that the opportunity-cost logic operates across governance types.

The aggregate consequence of this pattern is a substantial gap between where protection is concentrated and where conservation value is highest. Williams et al. (2022) find that the current global PA network is insufficient to safeguard half the world's mammal species from extinction because their ranges fall disproportionately outside the protected network. Chaplin-Kramer et al. (2023) show that only 15% of the "critical natural assets" — ecosystems providing 90% of nature's contributions to people globally, from freshwater to carbon storage — currently lie within formal protected areas.

Mouillot et al. (2024) synthesise these findings using a machine-learning approach, formalising a "socio-environmental niche" of protected areas: the multivariate space of geographic, climatic, and socioeconomic conditions characteristic of protected relative to unprotected land. Their results confirm that a compact set of covariates — remoteness, low agricultural suitability, proximity to existing PAs, and national income — explains a large share of the global distribution of PAs. This finding is a precondition for the approach taken in this thesis: if PA placement follows a systematic, learnable pattern in covariate space, then a supervised machine-learning model can be trained on historical designations and used to forecast where future designations will occur.

2.3 Machine Learning in Conservation: From Ecological Prediction to Policy Decisions

Machine learning has become a standard toolkit in quantitative conservation science. Most applications target biophysical outcomes: habitat suitability mapping from species-occurrence data, land-cover change detection from remote-sensing imagery, and deforestation-risk modelling from geospatial predictors. Mayfield et al. (2017) provide an influential demonstration that Random Forest classifiers (Breiman, 2001) trained on freely available satellite and topographic predictors match the accuracy of purpose-built deforestation models in Mexico and Madagascar. Their finding — that publicly available data are sufficient for high-quality land-change prediction — is directly relevant here, as the predictor set of this thesis relies entirely on such sources.

A methodologically distinct strand extends machine learning to *policy-driven* or institutional decisions rather than ecological outcomes. Greenhill et al. (2024) predict which rivers and wetlands fall under federal regulatory jurisdiction under the US Clean Water Act using a model trained on geospatial covariates. Their study demonstrates that machine learning can recover the implicit decision rules of government agencies from observed outcomes — a closely analogous task to predicting where governments will establish protected areas. Kleinberg et al. (2018) provide the theoretical underpinning: in many institutional prediction problems, models trained on historical decisions substantially outperform unaided human judgment by exploiting high-dimensional covariate patterns that are too complex for human experts to track consistently.

These studies jointly establish three conditions that motivate the approach of this thesis. First, government conservation decisions follow systematic spatial patterns in geographic covariate space. Second, those patterns are recoverable from historical designation records using standard geospatial predictors. Third, gradient-boosted tree models (Ke et al., 2017) are well suited for this task: they handle non-linear interactions among the 73 engineered covariates without requiring prior specification of functional forms, naturally manage the extreme class imbalance of rare transition events, and support post-hoc interpretation via SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017).

Despite this progress, existing machine-learning applications in conservation do not target the specific problem of predicting *near-term PA designation*. Most models are retrospective — classifying existing land cover or regulatory status — or focus on ecological rather than institutional outcomes. No prior study combines a temporal panel structure, multi-region comparison, probability calibration, and forward deployment into a coherent tool for PA transition-risk assessment. This thesis fills that gap.

2.4 The 30×30 Target, Economic Consequences, and Transition Risk

Reaching the 30×30 target by 2030 requires an unprecedented acceleration of PA establishment. Robinson et al. (2024) document that achieving 30% protection will require not only expanding conventional government-managed PAs but also incorporating "Nature's Strongholds" — large, intact wilderness areas — alongside novel governance modalities such as indigenous and community conserved areas and other effective area-based conservation measures (OECMs). Dasgupta et al. (2025) analyse implementation experiences across countries and find that designation pace is highly heterogeneous and constrained by land availability, political will, and financial resources; under current trajectories, most countries fall far short of the target.

The economic consequences of large-scale PA expansion are significant. PA designation imposes binding land-use restrictions that curtail agriculture, resource extraction, and infrastructure development. Reynaert et al. (2024) quantify the environmental and economic effects of PA policy and document substantial land-value and income effects for affected communities and landowners. These costs are spatially concentrated: they fall disproportionately on those whose assets sit in the predicted expansion zone. The inability to anticipate where the next wave of PAs will be established therefore generates material uncertainty for landowners, farmers, agribusinesses, and investors with land-linked exposures.

Emerging nature-related financial disclosure frameworks have begun to classify this uncertainty as a formal category of *transition risk* (Govind and Webb, 2025) — analogous to the stranded-asset risk created by carbon pricing in the climate-finance literature. Institutions with land-collateralised lending books, agricultural supply chains, or investments in resource-extraction concessions all face potential exposure to sudden PA designation, particularly as political urgency around 30×30 mounts. Despite this growing demand for spatially explicit transition-risk inputs, no prior academic study provides calibrated, forward-looking probability maps for near-term PA establishment that could be directly used in portfolio risk assessment. This thesis addresses that gap: by training a temporal machine-learning model on historical PA designation patterns, calibrating its outputs to interpretable probabilities, and deploying it forward to the 2025–2030 horizon, it provides a spatial risk tool directly relevant to nature-related financial disclosure.

3 Data

3.1 Study Areas

3.1.1 South America (Primary Model)

South America is selected as the primary study region for three reasons. First, it contains some of the world's most biodiverse and ecologically heterogeneous ecosystems, including the Amazon rainforest, the Cerrado savanna, the Andean highlands, and the Atlantic Forest, providing a diverse spatial canvas on which to train and evaluate the model. Second, protected-area coverage has expanded substantially since 2000 — from roughly 15% to 25.6% of terrestrial area — generating sufficient temporal variation in designations to train a transition model. Third, the continent encompasses wide variation in institutional capacity, governance quality, and economic development, enabling the model to capture a broad range of drivers.

The study covers all terrestrial land areas of South America at a 1×1 km resolution, yielding approximately 20.8 million unique grid cells observed annually from 2000 to 2024

(25 time periods), for a total of approximately 413 million pixel–year observations after excluding already-protected pixels.

3.1.2 United States (Robustness Check 1)

The United States provides a structurally different test environment: a federal institutional system with well-documented public land designations, high data quality, and a current PA coverage of only 7.7% — well below the 30% target. The large gap between current coverage and the 30×30 commitment implies that the US faces substantial future designation pressure, making it a policy-relevant case. Estimating a separate US model also tests whether the predictive framework generalises to a different governance context.

3.1.3 Southeast Asia (Robustness Check 2)

Southeast Asia is chosen as a second robustness region because it combines a high rate of ongoing deforestation with a relatively low current PA coverage of 14.0%. The region encompasses diverse governance regimes across ASEAN member states, from federated systems to centralised states, and faces acute pressure from agricultural expansion and extractive industries. This combination of high conservation urgency and distinct institutional context provides a challenging test of the model's cross-context generalisability.

3.2 Data

3.2.1 Target Variable

Protected-area boundaries and designation dates are obtained from the World Database on Protected Areas (WDPA; UNEP-WCMC and IUCN, 2024b,a), the most comprehensive global PA inventory. WDPA polygons are rasterised to the 1 km reference grid for each year of the panel: a cell is classified as protected in year t if any portion of the cell overlaps with a WDPA polygon whose designation year is $\leq t$. This cumulative definition ensures that once protected, a cell remains so — consistent with the transition formulation in Equation (1).

The WDPA records the initial transition into formal protected status. If a PA is subsequently de-listed or has its boundaries contracted — a relatively rare event in the data

— the cell retains protected status in our representation. This choice is conservative in the sense that it errs toward recording more protection, but it reflects the legal reality that formal PA de-listing is unusual and that transition-risk analysis is concerned with the risk of *designation*, not of de-listing.

A known limitation of the WDPA is that designation years can be imprecise: gazette dates may lag the formal legal decision by one to three years in some jurisdictions, or may be retrospectively entered into the database. This uncertainty is mitigated by using a five-year lookahead window, which reduces sensitivity to exact label dates.

3.2.2 Predictor Variables

Predictor variables are assembled to capture three broad categories of influence on PA designation: (i) existing conservation context, (ii) environmental and ecological conditions, and (iii) human pressure and economic activity. Table 1 summarises the full set of predictors, their sources, and temporal coverage.

Table 1: Predictor variables and data sources used in all three regional models.

Variable / Group	**Description**	**Source & Coverage**
Existing Conservation Context		
Distance to nearest PA	Euclidean distance to the boundary of any existing PA in year $t-1$. Captures spatial path dependence: new designations cluster near the current network.	WDPA (UNEP-WCMC and IUCN, 2024b); annual 2000–2024
Environmental and Ecological Conditions		
Vegetation index (NDVI)	Annual median Normalised Difference Vegetation Index; proxy for vegetation density and ecosystem productivity.	MODIS MOD13Q1 (Didan, 2021); annual 2000–2024

Variable / Group	Description	Source & Coverage
Deforestation / forest cover	Tree canopy cover (2000 baseline) and cumulative annual forest loss; captures habitat extent and land-use pressure.	Hansen Global Forest Change (Hansen et al., 2013); 2000–2024
Wildfire occurrence	Annual burned-area indicator; captures fire disturbance and ecosystem stress.	MODIS MCD64A1 (Giglio et al., 2021); annual 2000–2024
Elevation and slope	Topographic characteristics influencing accessibility, agricultural suitability, and remoteness.	CGIAR-CSI SRTM V4 (Jarvis et al., 2008); static
Climate variables	19 bioclimatic variables (temperature, precipitation, seasonality, extremes) characterising long-run environmental conditions.	WorldClim V1 (Hijmans et al., 2005); static (1950–2000 climatology)
Global Safety Net (GSN) layers	Conservation-priority indicators: high-biodiversity areas, intact wilderness, climate-stabilisation zones, wildlife corridors, and WWF terrestrial ecoregions.	Global Safety Net (Dinerstein et al., 2020); static (2020)
Land cover	Annual dominant land-cover class (IGBP scheme); distinguishes forest, cropland, urban, wetland, and other types.	MODIS MCD12Q1 (Friedl and Sulla-Menashe, 2022); annual 2001–2024
Human Pressure, Economic Activity, and Infrastructure		

Variable / Group	Description	Source & Coverage
Population density	Gridded population counts; proxy for development intensity and opportunity cost of protection.	GPW v4 (Center for International Earth Science Information Network (CIESIN), Columbia University, 2018); 5-year intervals 2000–2020
Night-time lights (HNTL)	Harmonised night-time light intensity (DMSP/VIIRS); proxy for economic activity and built infrastructure.	HNTL (Li et al., 2020); annual 2000–2024
Road infrastructure	Major road network density; captures accessibility and development pressure.	GRIP (Meijer et al., 2018); static (2018)
Energy and extractive infrastructure	Power plants and oil & gas well/pipeline locations as proxies for industrial capital investment and development pressure.	Global Power Plant Database (Global Energy Observatory et al., 2019); Oil & Gas Infrastructure (Omara et al., 2023); static
Indigenous territories	Distance to boundaries of formally recognised indigenous and community lands; captures institutional overlap between conservation and indigenous rights.	LandMark (LandMark – Global Platform of Indigenous and Community Lands, 2023); static
Agricultural land value	Country-level FAO gross production value spatially allocated to agricultural pixels (cropland and pasture) using MODIS land-cover weights; proxies the economic opportunity cost of protecting productive land.	FAO FAOSTAT (Food and Agriculture Organization of the United Nations, 2024); annual 2000–2024
Governance and Institutional Context		

Variable / Group	Description	Source & Coverage
Government effectiveness & rule of law	WGI Government Effectiveness and Rule of Law scores; country-level measures of institutional capacity to translate policy commitments into on-the-ground designations.	World Bank WGI (World Bank, 2025); annual 2000–2024
Democratic governance	V-Dem Liberal Democracy Index; captures the quality of democratic institutions and civil society, which shape conservation policy responsiveness.	V-Dem v15 (V-Dem Institute, 2025); annual 2000–2024
Executive political ideology	DPI executive ideology score (left/centre/right); controls for political-economy variation in conservation priorities across governments.	DPI 2020 (Scartascini et al., 2021); annual 2000–2020

Existing conservation context. Distance to the nearest PA boundary (`dist_wdpa`) captures spatial path dependence: designations cluster near the existing network as buffer zones and corridor links. Lagging by one year prevents the model from exploiting same-year PA information.

Environmental conditions. NDVI, deforestation, wildfire, elevation, slope, and WorldClim bioclimatic variables characterise biophysical suitability and ecological value. The GSN layers add globally standardised conservation-priority scores covering biodiversity hotspots, intact wilderness, and climate-stabilisation areas.

Human pressure and economic activity. Population density, night-time lights, and distances to roads, power plants, and oil and gas installations characterise the intensity of human land use and the opportunity cost of designation. Agricultural land value —

computed by spatially allocating country-level FAO gross production value across cropland and pasture pixels using MODIS land-cover weights — provides a direct economic proxy: high-value agricultural land is more costly to designate and thus faces stronger political resistance to protection.

Institutional predictors. Distance to formally recognised indigenous and community lands (LandMark) captures the spatial overlap between indigenous territories and conservation expansion, a substantively important pattern in South America and Southeast Asia. Country-level governance indicators — WGI Government Effectiveness, WGI Rule of Law, V-Dem Liberal Democracy Index, and DPI executive political ideology — capture the institutional capacity and political economy of each country. Empirically, designation pace and spatial pattern vary substantially with governance quality: countries with strong rule of law and effective institutions are better able to translate international commitments into formal designations.

Static versus dynamic predictors. Variables that are time-invariant — elevation, slope, WorldClim climatology, road networks, GSN layers, oil and gas infrastructure, and distance to indigenous territories — are treated as static and repeated across all years for a given cell. Time-varying variables — NDVI, deforestation, wildfire, population density, night-time lights, distance to PAs, agricultural land value (annual FAO production), and governance indicators (WGI, V-Dem, DPI; assigned by country-year) — are updated annually.

Missing values. Missing values arise from genuine data gaps (e.g. incomplete satellite coverage in cloud-persistent regions). Rather than imputing missing values, they are retained as `NA` and handled natively by the tree-based models used in this study, which accommodate missingness through their internal split logic without distorting the feature distribution. For population density (available at five-year intervals) and night-time lights (small end-of-series gap), limited forward-filling within each cell is applied to ensure annual coverage.

3.3 Harmonisation and Panel Construction

Reference grid and coordinate system. All analyses are conducted on a regular 1 × 1 km reference grid defined in the coordinate reference system EPSG:3857 (WGS 84 Web Mercator). The 1 km resolution reflects a trade-off between spatial detail and data compatibility: the majority of key predictors (population density, climate variables, long-run infrastructure layers) are available only at spatial scales of 1 km or coarser, so using a finer grid would require extensive interpolation of the most policy-relevant features without improving information content. While EPSG:3857 introduces areal distortions at high latitudes, these distortions are uniform across all inputs within each study region and do not affect the relative rankings produced by the model.

A fixed land-only backbone grid defines the universe of cells eligible for analysis. Ocean pixels and pixels outside each study region are excluded uniformly from all datasets.

Reprojection and resampling. All input rasters are harmonised to the backbone grid in terms of coordinate reference system, resolution, and spatial extent. Continuous variables (elevation, NDVI, population density, climate, night-time lights) are resampled using bilinear interpolation, which reduces aliasing artefacts relative to nearest-neighbour resampling. Categorical and binary layers (land-cover classes, PA status, conservation priority indicators) are resampled using nearest-neighbour interpolation to preserve discrete class boundaries. For image collections with multiple observations per year, a single annual raster is constructed by taking the annual median for continuous variables and aggregating fire events cumulatively.

Panel construction. Following spatial and temporal harmonisation, all layers are merged into a unified pixel–year panel. Each row corresponds to a single grid cell i in a specific year t and contains: (i) the binary target variable transition$_{i,t}$; (ii) the full set of static and dynamic predictor variables; and (iii) cell and year identifiers. The resulting South America panel contains approximately 413 million pixel–year observations across 20.8 million unique grid cells observed over 25 annual periods (2000–2024). Comparable panels are constructed for the United States and Southeast Asia.

4 Methods

Figure 1 gives an overview of the end-to-end pipeline. The subsections below describe each stage in detail.

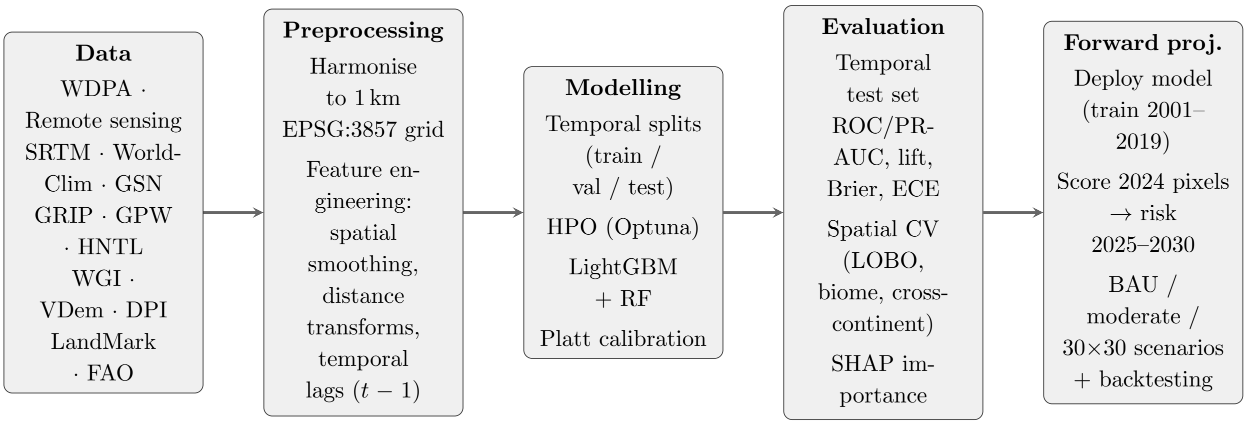

Figure 1: End-to-end pipeline overview. Data from multiple source categories are harmonised and merged into a pixel–year panel; LightGBM and Random Forest models are trained on 2000–2013, calibrated, and evaluated using temporal and spatial cross-validation; a deployment model trained on 2001–2019 generates forward-looking transition-risk maps to 2030 under three scenarios.

4.1 Modelling Framework

Problem formulation. The objective is to estimate, for each currently unprotected 1 km^2 grid cell i observed in year t, the probability that cell i becomes newly designated as a protected area within the subsequent five years. Let $y_{i,t} \in \{0, 1\}$ denote whether cell i is protected in year t. The target variable is a five-year forward-looking binary indicator:

$$\text{transition}_{i,t} = \begin{cases} 1 & \text{if } y_{i,t} = 0 \text{ and } y_{i,t+k} = 1 \text{ for any } k \in \{1, \dots, 5\}, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

A five-year horizon is chosen because it corresponds to the medium-term planning cycle of most conservation programmes and aligns with the 2025–2030 window of the 30×30 target.

Risk set and right-censoring. Only unprotected pixels ($y_{i,t} = 0$) are eligible for the transition and therefore included in the analysis. Once a pixel becomes protected, it exits the *risk set* permanently and does not contribute further observations. This prevents the model from learning to predict non-events for land that is already protected.

Right-censoring is a technical constraint arising from the finite data horizon. To observe the full five-year outcome window, the last year for which a label can be constructed is 2019 (using PA designations through 2024). All pixel–year observations beyond 2019 are excluded from model training and evaluation; the target variable is undefined for them because we cannot verify whether a transition occurs in the full subsequent five-year window. This cutoff year is denoted `LAST_LABEL_YEAR` = 2019.

Rare-event structure. Transitions into protected status are rare: fewer than 0.5% of eligible pixel–year observations record a positive outcome. This extreme class imbalance is not an artefact of data collection but reflects the true pace of PA expansion relative to the vast unprotected land area. Standard classifiers trained on such data tend to predict the majority class (non-transition) for almost all observations and achieve high nominal accuracy while being useless for identifying future designations. The modelling choices described in Sections 4.3–4.5 are designed specifically to address this challenge.

Relation to discrete-time hazard models. The panel formulation of Equation (1) is conceptually related to discrete-time hazard (or duration) models used in econometrics to study the timing of events such as unemployment spells, technology adoption, or firm bankruptcy (Allison, 1982). In the hazard interpretation, each unprotected pixel is "at risk" of designation in each year, and the model estimates the conditional probability of transitioning given that no transition has occurred yet. Unlike parametric hazard models, the tree-based machine-learning approach used here imposes no functional form on the relationship between predictors and the transition probability, and handles complex interactions among the 73 engineered covariates without prior specification.

Prediction, not prescription. The framework learns *where governments have historically designated* PAs and extrapolates this pattern forward. It does not predict where PAs *should* be designated for maximum conservation impact — that is the domain of reserve-selection algorithms. The two questions have different answers because real designation is shaped by political feasibility and opportunity costs, not biodiversity optimisation (Schuster et al., 2023). The gap between predicted and optimal designations is itself a substantive finding discussed in Section 6.2.

4.2 Feature Engineering

Raw cell-level features describe conditions at a single pixel. Because PA designation is influenced by conditions in the surrounding landscape — a pixel embedded in an agricultural frontier faces different designation pressure than an isolated clearing within a large forest block — additional features are engineered to encode spatial context at multiple scales.

Multi-scale spatial smoothing. For a selected subset of time-varying predictors (NDVI, population density, night-time lights, deforestation intensity, and wildfire), uniform averaging kernels of size 4×4, 16×16, and 64×64 grid cells are applied, corresponding to neighbourhood extents of 4 km, 16 km, and 64 km respectively. These three scales are designed to capture: (i) immediate local context (4 km: adjacent land use, buffer zones); (ii) sub-regional landscape dynamics (16 km: agricultural frontier gradients, infrastructure corridors); and (iii) landscape-scale patterns (64 km: macro-regional biome boundaries, large-scale deforestation fronts). Including all three scales allows the model to select the most informative spatial context for each combination of predictor and location without imposing a single fixed neighbourhood size.

Distance-based features. In addition to smoothed averages, Euclidean distances to the nearest PA boundary, road segment, power plant, and oil and gas installation are included as features. These distance measures directly encode proximity to conservation and development infrastructure, capturing processes such as incremental PA network expansion and the deterrent effect of industrial activity on designation.

Temporal lag. All time-varying features — including the distance to existing PAs — are lagged by one year (features at year t reflect conditions at year $t-1$). This ensures that the model cannot use information about PAs designated in year t itself to predict the transition target for year t, which would constitute temporal leakage.

4.3 Machine Learning Models

Three modelling approaches are estimated for each region, in increasing order of complexity.

Static Suitability Baseline. The static model is estimated on cross-sectional data from the year 2000 only. It learns to distinguish pixels that are already protected from those that are not, using only the time-invariant geographic characteristics available at baseline. Suitability scores for pixels unprotected in 2000 are then compared to PA designations observed between 2000 and 2024. This model captures long-run spatial correlates of protection (remoteness, elevation, biodiversity value) but ignores all temporal dynamics. It establishes a lower bound on predictive performance: any improvement over the static baseline reflects the informational value of temporal variation in dynamic predictors and the transition-specific training target.

Random Forest. A Random Forest (Breiman, 2001) classifier is trained on the pixel–year panel. A Random Forest constructs a large number of decision trees (here: 300 trees), each trained on a bootstrap sample of the data with a random subset of features considered at each split. Final predictions are averaged across all trees, which reduces variance relative to any individual tree and provides a built-in measure of feature importance through the average reduction in node impurity. For example, one tree might learn that pixels with `dist_wdpa` < 5 km *and* GSN biodiversity score > 0.6 have a $3\times$ higher transition rate than average; across 300 such trees, these local rules average out to a stable probability estimate that is robust to any one tree's idiosyncrasies. Class imbalance is addressed via `class_weight="balanced_subsample"`, which reweights training examples within each tree so that minority-class (transition) observations receive weight inversely proportional to their frequency. The Random Forest serves as a nonparametric benchmark and provides complementary feature importance diagnostics.

LightGBM. Light Gradient-Boosting Machine (LightGBM; Ke et al., 2017) is the primary production model. Gradient boosting builds an ensemble of decision trees sequentially: each new tree is fitted to the residuals of the current ensemble, so successive trees correct the errors of their predecessors. LightGBM introduces two computational optimisations — Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) — that make it substantially faster than standard gradient boosting on large datasets without loss of accuracy. Class imbalance is addressed via the `scale_pos_weight` parameter, which is set to the ratio of negative to positive observations in the training data:

$$\texttt{scale\_pos\_weight} = \frac{N_{\text{neg}}}{N_{\text{pos}}}. \tag{2}$$

This is equivalent to upweighting the minority class in the loss function, steering the boosting procedure toward correct identification of rare transitions.

Table 2 reports the key hyperparameter values for both models. Hyperparameters were selected via temporal cross-validation optimising Precision–Recall AUC on a held-out validation period (see Section 4.5).

Table 2: Key hyperparameter values for the Random Forest (RF) and LightGBM (LGBM) transition models.

Parameter	**RF**	**LightGBM**	**Notes**
Number of trees / rounds	300	Tuned per region	Fixed from CV
Max depth	Unlimited	Tuned	LGBM: -1 = no limit
Min samples per leaf	1	Tuned	LGBM: `min_child_samples`
Class imbalance	`balanced_subsample`	`scale_pos_weight`	See Eq. (2)
Random seed	42	42	Fixed for reproducibility

4.4 Probability Calibration

Raw model outputs — the predicted class probabilities from LightGBM or Random Forest — are scores, not calibrated probabilities. A score of 0.7 does not mean the pixel has a 70% chance of being designated; it means only that the pixel ranks higher than most others. This distinction matters for transition-risk applications, where analysts need to know not just the rank order of risky pixels but also the magnitude of the risk. For example, a portfolio manager screening land-secured loans wants to flag all loans where the underlying collateral has a designation probability above a chosen threshold (say, 2%), not merely the top-ranked parcels.

The mismatch between raw scores and true probabilities arises partly because class weighting (Equation 2) modifies the effective class prior seen by the model: the training data that reach each tree have been rebalanced, so the model learns to predict relative risk within the rebalanced distribution rather than within the true distribution.

Calibration is performed using Platt scaling (Platt, 1999): a logistic regression model is fitted to the logit-transformed out-of-fold predictions from temporal cross-validation, with the true binary labels as the outcome. This fitted logistic function maps raw scores to calibrated probabilities. For example, a pixel assigned a raw LightGBM score of 0.85 might receive a calibrated probability of only 0.62 after Platt scaling, because the class-weighting procedure (Equation 2) causes the model to systematically over-rank in the high-score range; the logistic recalibration corrects this inflation to align predictions with observed transition rates. Isotonic regression is used as a robustness check. The calibrators are fitted exclusively on out-of-fold cross-validation predictions, ensuring that no information from the test period enters the calibration. Calibrated probabilities are then clipped to the interval $(10^{-6}, 1 - 10^{-6})$ for numerical stability. All probability maps and risk scores reported in this thesis are based on calibrated LightGBM predictions.

4.5 Training, Validation, and Test Design

Temporal splits. Model training and evaluation strictly preserve temporal order to ensure that the model is evaluated on its ability to forecast *future* designations using only *past* information. The panel is divided into three non-overlapping periods:

- **Training period:** 2000–2013. The model learns the relationship between pixel characteristics and the five-year transition probability.
- **Validation period:** 2014–2016. Used for hyperparameter selection and early stopping. Not used for final model training or for reported evaluation metrics.
- **Test period:** 2017–2019. Held out entirely and used only for final evaluation. The last year, 2019, corresponds to `LAST_LABEL_YEAR`; observations from 2020 onwards are right-censored and excluded.

This design means that every metric reported in Section 5 is an out-of-sample estimate of future-period performance: the model was trained on 2000–2013 and evaluated on 2017–2019, with no data from 2014 onwards influencing the training parameters.

Leakage prevention. Multiple safeguards prevent future information from reaching the model. (i) All time-varying features are lagged by one year, so features at year t reflect conditions at year $t-1$. (ii) Distance to existing PAs is computed using only PAs designated up to year $t-1$, preventing the model from indirectly inferring same-year designations. (iii) Hyperparameters are tuned on the validation period, which strictly precedes the test period. (iv) Probability calibrators are fitted on out-of-fold cross-validation predictions, not on test data.

Temporal sample weighting. Because conservation-policy dynamics evolve over time — the post-2010 acceleration toward the Aichi targets and the subsequent Kunming-Montreal framework have altered designation patterns — observations in later years within the training period are assigned higher weights, with a linear ramp from 1.0 in 2000 to a user-specified maximum in 2013. This places greater emphasis on recent patterns without discarding the earlier data.

Class imbalance handling. Transitions are rare (≈ 0.3–0.5% of eligible pixel–year observations). No oversampling or undersampling is applied; the empirical frequency is preserved. Instead, class weights (LightGBM: `scale_pos_weight`; Random Forest: `balanced_subsample`) adjust the contribution of minority-class observations to the loss function. This approach avoids distorting the spatial or temporal structure of the data.

Reproducibility. All random components (tree construction, bootstrap sampling) are controlled via fixed random seeds (seed 42 throughout). The full pipeline is implemented in Python using LightGBM (Ke et al., 2017), scikit-learn (Pedregosa et al., 2011), and GeoPandas, with intermediate outputs stored in Parquet format. Large-scale training is executed on the ETH Zurich Euler HPC cluster using SLURM job arrays.

4.6 Evaluation Metrics

Because the positive class constitutes fewer than 0.5% of observations, accuracy-based metrics (e.g. overall accuracy, F_1-score) are uninformative: a model that predicts zero transitions everywhere achieves $> 99.5\%$ accuracy (Davis and Goadrich, 2006). Evaluation therefore focuses on metrics that are robust to severe class imbalance and that directly reflect the ranking quality of predicted risks.

Precision–Recall AUC (PR-AUC). The primary metric. The precision–recall curve plots precision (fraction of top-ranked pixels that actually transition) against recall (fraction of all transitions captured) as the classification threshold is varied:

$$\text{Precision}(k) = \frac{TP(k)}{TP(k) + FP(k)}, \quad \text{Recall}(k) = \frac{TP(k)}{TP(k) + FN(k)}. \tag{3}$$

PR-AUC is the area under this curve, integrated across all thresholds. Under severe class imbalance, PR-AUC is a more informative summary of discriminative performance than ROC-AUC because it places weight on the model's ability to correctly rank positive observations at the top of the score distribution (Davis and Goadrich, 2006).

ROC-AUC. Reported as a complementary metric to facilitate comparison with existing studies. ROC-AUC measures the probability that a randomly chosen positive observation receives a higher predicted score than a randomly chosen negative observation. While less sensitive than PR-AUC under extreme imbalance, it provides a useful check for degenerate or anti-predictive models.

Precision at top-k% and lift. For policy use, the most relevant question is: "if I inspect the top k% of predicted-risk pixels, what fraction will actually be designated?" This is precision at top k% of the score distribution ($P@k\%$). To account for differences in base rates across evaluation splits, we also report *lift*:

$$\text{Lift}@k\% = \frac{P@k\%}{\text{baseline positive rate}}. \tag{4}$$

A lift of 60 at the top 1% means that inspecting the top 1% of pixels yields 60 times more future designations per unit area than random selection. We report lift at $k \in \{1, 5, 10\}$.

Brier score. The mean squared error of calibrated predicted probabilities:

$$\text{Brier} = \frac{1}{N} \sum_{i,t} \left(\hat{p}_{i,t} - y_{i,t} \right)^2. \tag{5}$$

A lower Brier score indicates better-calibrated and more accurate probability estimates. The Brier score is bounded between 0 (perfect) and 1 (maximally wrong). Under a positive rate of π, a naive constant-prediction model predicts $\hat{p} = \pi$ everywhere and achieves a Brier score of $\pi(1 - \pi)$.

Expected Calibration Error (ECE). ECE measures the average discrepancy between predicted probabilities and observed frequencies, computed by binning predictions into equal-width intervals and comparing the average predicted probability to the observed transition rate within each bin:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|n_b|}{N} \left| \overline{p}_b - \overline{y}_b \right|, \tag{6}$$

where n_b is the number of observations in bin b, $\overline{p}_b$ is their mean predicted probability, and $\overline{y}_b$ is their observed transition rate. ECE is reported before and after Platt scaling to quantify the calibration improvement.

4.7 Spatial Generalisation Framework

In addition to temporal out-of-sample evaluation, spatial robustness is assessed through a three-layer framework that tests whether the model generalises to unseen geographies within and across continents.

4.7.1 Layer 1 — Leave-One-Biome-Out (LOBO)

Biomes are defined using the Global Safety Net (GSN) terrestrial ecoregion layer, which partitions each continent into WWF biome classes. For each biome b, a separate model is trained on all observations from the other biomes and evaluated on the held-out biome's test-period observations. This tests whether the model captures continent-wide patterns that transfer to geographic areas excluded from training — the strictest within-continent generalisation test. Per-biome ROC-AUC is reported; low LOBO performance in a biome indicates that designation patterns there are structurally different from the rest of the region.

4.7.2 Layer 2 — Biome-Stratified Evaluation

The production model — trained on all biomes simultaneously — is scored on the test period and evaluation metrics are disaggregated by biome. This diagnostic reveals which biomes the full model handles well versus poorly when trained on all available information. Unlike LOBO, it does not test cross-biome generalisation but provides a useful characterisation of within-model performance heterogeneity.

4.7.3 Layer 3 — Cross-Continental Transfer

The South America production model is scored on US test observations and vice versa, using the common set of 73 features available in both regions. If the model generalises cross-continentally, it has learned universal ecological and geographical drivers of PA designation. If transfer performance is near random, it implies that designation patterns are continent-specific — a substantive finding about the political economy of conservation.

4.8 Forward Prediction to 2030

Deployment model. To generate forward-looking probability maps for 2025–2030, a deployment model is trained on the full labelled window 2001–2019 (all years for which the transition target can be constructed without right-censoring), using the hyperparameters fixed by the evaluation stage. The deployment model is calibrated on out-of-fold predictions from this extended training period.

Forward inference. Inference is performed on the 2024 covariate snapshot for all pixels unprotected as of 2023. The deployment model assigns each candidate pixel a calibrated probability of becoming protected between 2025 and 2030. This constitutes a strict temporal out-of-sample forecast: no 2024 information enters model training.

Policy scenarios. Three scenarios are derived from the resulting probability surface by varying the cumulative area designated:

- **Business-as-usual (BAU):** cumulative new area equals the observed 2019–2024 designation volume, projecting the recent empirical designation rate forward.
- **Moderate:** cumulative area targets the midpoint between each region's current coverage and 30%, applied proportionally.
- **30×30:** cumulative area targets full 30% terrestrial coverage by 2030.

Under each scenario, the model selects the highest-probability pixels to fill the required area, producing spatially explicit designation maps. Hotspot clusters are identified via DBSCAN spatial clustering (Ester et al., 1996) on the top-probability pixels.

Backtesting. Methodological validity is established by backtesting: pseudo-forecasts are generated at historical origins $T \in \{2009, 2011\}$, each time training on 2001–$(T-1)$, scoring unprotected pixels in year T, and evaluating against realised designations in years $T+1$ through $T+5$. This walk-forward evaluation empirically verifies that the forward deployment procedure generalises to out-of-sample designation events.

5 Results

5.1 Descriptive Statistics and Transition Dynamics

The South America panel contains approximately 413 million pixel–year observations spanning 20.8 million unique 1 km^2 grid cells over 25 annual periods (2000–2024). By 2024, 25.6% of South America's terrestrial area is formally protected, up from approximately 15% in 2000. As a consequence of this cumulative expansion, the risk set shrinks from roughly 20.8 million eligible pixels in 2000 to approximately 15.7 million by the end of the sample.

Transitions from unprotected to protected status are rare events. The average transition rate across the full South America panel is approximately 0.5–0.8% per pixel–year, implying that a given pixel has less than a 1-in-100 chance of being designated in any given year. This rate is not constant over time: designation rates were higher during 2000–2013, a period of active PA expansion in the Amazon and Cerrado, and declined thereafter. In the held-out test period (2017–2019), 380,064 transitions occur among approximately 47.7 million eligible pixel–year observations, corresponding to a positive rate of approximately 0.8%.

The United States panel covers approximately 200 million pixel–year observations. Current protection stands at 7.7%, and the test-period transition rate is approximately 0.04% — an order of magnitude lower than South America — reflecting the US's severely constrained designation pace relative to its 30% target. Southeast Asia covers approximately 100 million pixel–year observations, with current PA coverage of 14.0% and an intermediate test-period transition rate.

These descriptive patterns highlight three defining features of the data that shape all modelling choices: (i) a large risk set; (ii) strong temporal heterogeneity in transition rates; and (iii) extreme class imbalance, particularly in the US and in later sample years.

5.2 South America: Model Performance

5.2.1 Discrimination Metrics

Table 3 reports temporal test-set performance for the LightGBM and Random Forest transition models applied to South America (test period: 2017–2019).

Table 3: Predictive performance on the temporal test set (2017–2019) for South America. Baseline positive rate: $\approx 0.80\%$.

Metric	LightGBM	Random Forest
ROC-AUC	0.897	0.927
PR-AUC	0.645	0.634
Precision @ top 1%	0.532	0.539
Lift @ top 1%	66.8×	67.6×
Precision @ top 5%	0.116	0.121
Lift @ top 5%	14.5×	15.2×
Brier score	0.00421	0.00458
ECE (post-calibration)	0.0081	0.0109

The LightGBM model achieves strong discrimination, with a ROC-AUC of 0.897 and a PR-AUC of 0.645. Given the positive rate of approximately 0.8%, the PR-AUC of 0.645

represents a substantially better ranking than a random classifier, which would achieve a PR-AUC equal to the positive rate (0.008). Focusing on the most policy-relevant threshold, the model attains a lift of 66.8× at the top 1% of predicted risk: screening the highest-risk 1% of pixels yields 66 times as many future designations per unit area as random selection. The Brier score of 0.00421 and ECE of 0.0081 confirm that calibrated probabilities are well-aligned with realised transition rates.

Although the Random Forest achieves a higher ROC-AUC (0.927 vs. 0.897), LightGBM is retained as the primary model throughout the analysis because it attains a superior PR-AUC (0.645 vs. 0.634) and a lower Brier score (0.00421 vs. 0.00458) and ECE (0.0081 vs. 0.0109). Under extreme class imbalance, PR-AUC is the more informative discrimination metric — the ROC curve is dominated by the majority class and can appear strong even when precision at actionable thresholds is poor. The LightGBM advantage in calibrated probability accuracy is also practically important: it produces tighter, more reliable risk scores for portfolio screening.

5.2.2 Event-Level Coverage

Pixel–year metrics penalise a model that correctly identifies *where* a designation will occur but not its exact year. An operationally relevant complement is the *event-level* question: across all test-period designations, what fraction are captured by the top-1% risk zone at any point in the test window?

Under event-level aggregation, the LightGBM model captures 90.4% of all test-period protected-area establishments within the top 1% of predicted risk. Put differently, screening the highest-risk 1% of unprotected pixels — approximately 156,000 km^2 out of 15.7 million km^2 eligible — would have flagged 9 out of every 10 actual new PAs before designation. This result confirms that the moderate pixel–year PR-AUC does not reflect poor spatial targeting but rather the difficulty of predicting the exact year of a designation, which is subject to political and administrative noise.

5.2.3 Probability Calibration

Before Platt scaling, raw LightGBM scores are systematically overconfident in the high-probability range: the model assigns scores of 0.9 to observations that become protected only 60% of the time. After calibration, the ECE drops to 0.0081, confirming that calibrated probabilities closely track observed transition rates across all bins of the score distribution. Reliability diagrams before and after calibration are provided in Appendix C.

The practical implication is that a pixel with a calibrated predicted probability of 0.03 has a 3% chance of being designated within five years — a statement that is meaningful for portfolio screening and risk quantification, not merely for ranking.

5.2.4 Feature Importance (SHAP)

Figure 2 shows the SHAP summary plot for the LightGBM model, ranking predictors by their mean absolute contribution to the log-odds of designation.

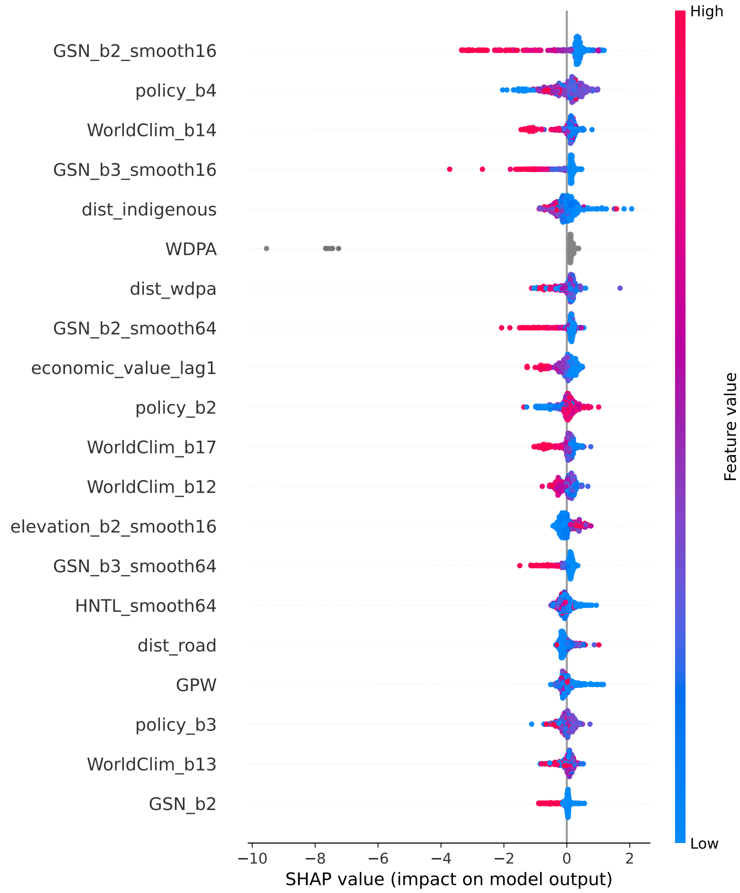

Figure 2: SHAP feature importance for the South America LightGBM model. Variables are ordered by mean absolute SHAP value. Horizontal spread indicates the range of SHAP contributions; colour indicates the raw feature value (red = high, blue = low).

The mean absolute SHAP values reveal four primary driver clusters. First, **conservation priority**: GSN biodiversity layers (principally `GSN_b2_smooth16` and `GSN_b3_smooth16`) rank as the single most important predictor group, with SHAP magnitudes well above all other categories. Pixels in globally significant biodiversity areas consistently attract designation — consistent with the ecological-value channel in Mouillot et al. (2024). Second, **governance quality**: WGI Rule of Law (`policy_b4`) is the second-most important individual feature, with V-Dem Liberal Democracy Index (`policy_b2`) and WGI Government Effectiveness (`policy_b3`) also in the top twenty. Countries with strong institutions translate international commitments into formal designations at higher rates, independently of ecological conditions. Third, **economic opportunity cost**: agricultural land value (`economic_value_lag1`) and distance to indigenous territories (`dist_indigenous`) are mid-tier predictors, with high agricultural value suppressing designation — the classic opportunity-cost mechanism of Joppa and Pfaff (2009). Infrastructure proximity (night-time lights, roads, population) enters at a lower tier, confirming that economic activity deters designation but that governance and conservation priority are stronger first-order determinants. Fourth, **climate and path dependence**: WorldClim bioclimatic variables and distance to the existing PA network (`dist_wdpa`) contribute at broadly similar magnitudes, confirming that both climatic suitability and spatial path dependence shape designation location without dominating the signal.

This driver hierarchy has a direct policy implication: historical PA designation is governed primarily by biodiversity value and institutional capacity, not merely by the economic feasibility of protection. Areas with high conservation priority but weak governance — and areas where strong governance exists but conservation priority is low — represent distinct policy challenges for achieving the 30×30 target.

5.2.5 Spatial Risk Maps

Figure 3 maps the top 1% of predicted pixels across South America, overlaid with observed PA designations during the test period (2017–2019). High-risk pixels form spatially coherent clusters rather than a diffuse pattern, concentrated in the Amazon basin periphery, the Cerrado–Amazon transition zone, the Andean foothills, and parts of Chile's southern regions. The spatial overlap between predicted high-risk zones and realised designations is visually strong and quantified by the 90.4% event-level recall reported above.

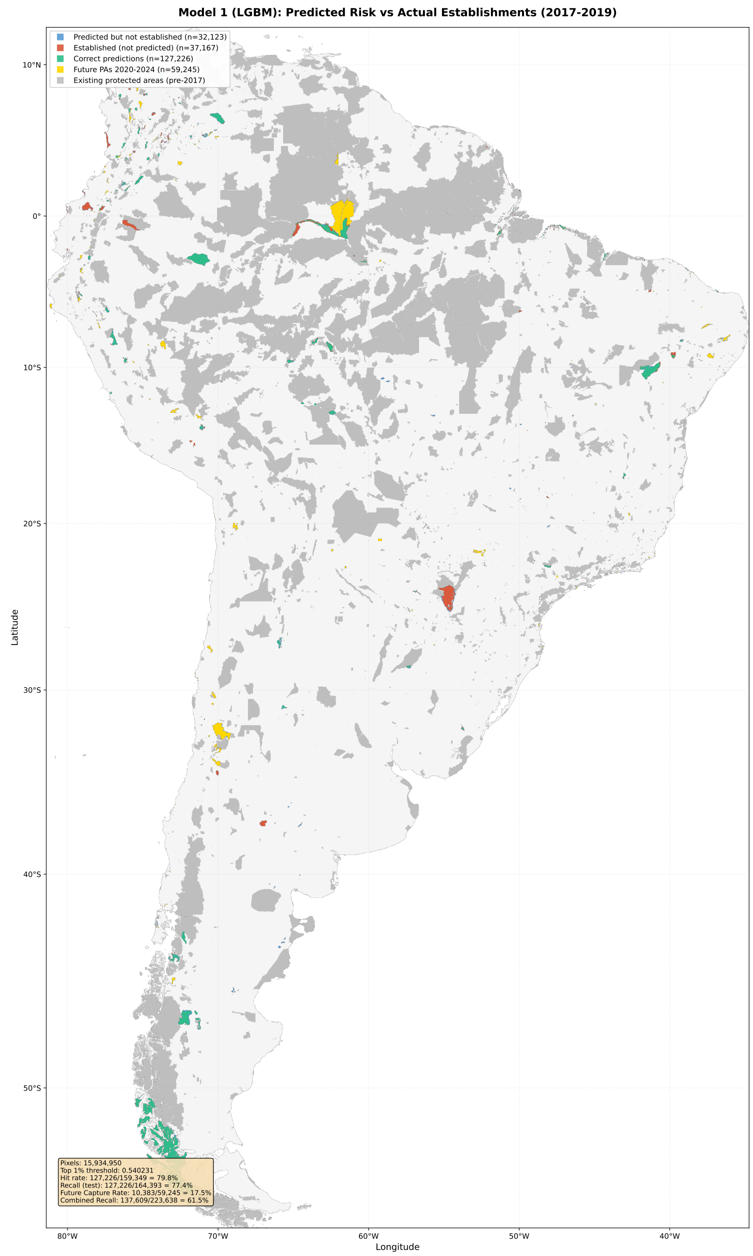

Figure 3: South America risk map: top 1% of predicted pixels (LightGBM). Grey areas indicate the top-1% risk zone; overlaid coloured symbols mark observed PA establishments during the test period (2017–2019).

The continuous probability map (Figure 4) shows a strongly right-skewed distribution: the vast majority of pixels receive near-zero calibrated probability, while a small upper tail concentrates substantially elevated risk. The spatial autocorrelation of high-risk zones confirms that the model captures landscape-scale processes rather than pixel-level noise.

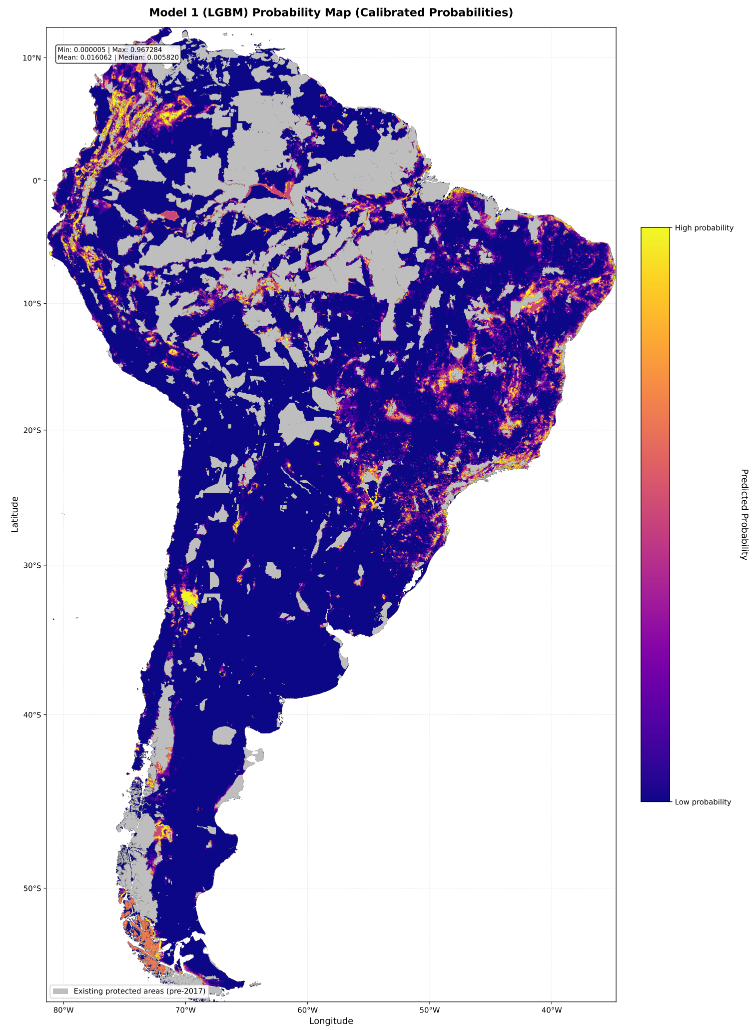

Figure 4: Continuous calibrated transition probabilities from the South America LightGBM model. Probabilities represent the estimated five-year designation likelihood for each unprotected pixel, conditional on 2019 covariate values.

5.2.6 Spatial Generalisation (LOBO)

The Leave-One-Biome-Out (LOBO) evaluation tests whether patterns learned on one set of biomes generalise to a held-out biome. The mean LOBO ROC-AUC across all nine South American biomes is 0.81, indicating substantial cross-biome transferability. However, performance varies markedly across biomes (Figure 5). Deserts and Xeric Shrublands (principally the Atacama) exhibit near-perfect LOBO accuracy (ROC-AUC 0.995), because PA designation there follows highly distinctive patterns — extreme remoteness, mineral concession boundaries — that are well predicted by the available feature set. By contrast, Temperate Broadleaf and Mixed Forests (Patagonia) achieve near-random LOBO performance (ROC-

AUC 0.491), suggesting that Chilean Patagonia follows political and ecological dynamics orthogonal to the rest of the continent: national park legislation driven by tourism value and international conservation pressure, not by the accessibility and land-use gradients that dominate elsewhere. Mediterranean Forests, Woodlands and Scrub (central Chile) also perform relatively weakly (ROC-AUC 0.701), reflecting low positive rates and spatially idiosyncratic designation processes.

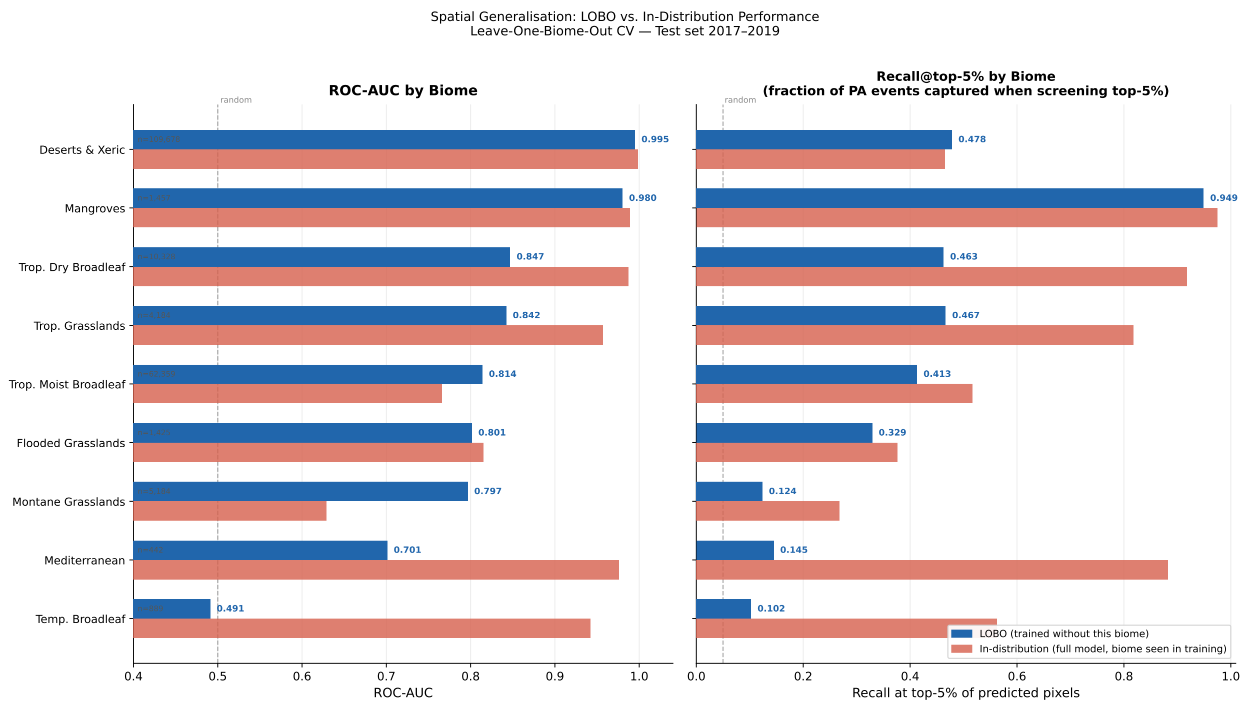

Figure 5: Per-biome LOBO ROC-AUC for South America (LightGBM). The horizontal dashed line marks random-classifier performance (0.5). Bars below the dashed line indicate biomes where the model trained on other biomes is anti-predictive for the held-out biome.

5.3 United States: Robustness Check

The US results confirm that the predictive framework extends to a structurally different institutional context. The overall LightGBM ROC-AUC on the US test set is 0.944, indicating strong overall rank discrimination. However, the PR-AUC is only 0.082, reflecting the extreme class imbalance in the US test set ($\approx 0.04\%$ positive rate — 20 times rarer than South America). The high ROC-AUC alongside the very low PR-AUC reflects the known divergence of these two metrics under extreme imbalance (Davis and Goadrich, 2006): the model ranks positives above negatives in aggregate but struggles to maintain high precision at any fixed threshold. The Lift at top 1% remains 80.6×, confirming that the top-ranked pixels are genuinely enriched for future designations even when the absolute precision is low.

A striking feature of the US data is the geographic concentration of designations: approximately 85% of all test-period PA establishments fall within the Deserts and Xeric Shrublands biome (Bureau of Land Management public land in the American West). This concentration simplifies the prediction problem in one sense — any model that learns to prioritise arid BLM land will score well overall — while creating severe imbalance for all other biomes.

The LOBO analysis reveals genuine geographic heterogeneity, with a mean LOBO ROC-AUC of 0.51 across the twelve US biomes. Temperate Conifer Forests (LOBO ROC-AUC 0.334) and Deserts and Xeric Shrublands (0.387) are the weakest biomes, performing at near-random or anti-predictive levels when held out from training. These represent cases where the mix of federal, state, and private land tenure — the main institutional driver of designation in these biomes — is not adequately proxied by the available geospatial covariates, which do not include land-tenure information. This is a genuine ecological and institutional finding, not merely a modelling artefact: PA designation in Pacific Northwest conifer forests follows complex multi-stakeholder processes that the model cannot capture from publicly available remotely-sensed data alone.

Full metrics tables for the US are provided in Appendix B.

5.4 Southeast Asia: Robustness Check

The Southeast Asia model achieves a ROC-AUC of 0.994 on the test set, with a PR-AUC of 0.919. These high values reflect the geographic concentration of PA designation in the region: new PAs are predominantly located in forested upland areas of Borneo, Sumatra, peninsular Malaysia, and the Philippines, which differ sharply from surrounding lowland agricultural zones in terms of elevation, canopy cover, and remoteness. The strong discrimination is therefore partly an artefact of this extreme spatial segregation — the same mechanism that produces perfect LOBO scores in South America's Atacama Desert. Nonetheless, the model is operationally useful: the Lift at top 1% is 94.2×, confirming that the top-ranked pixels are genuinely enriched for future designations. Full metrics are provided in Appendix B.

The nine LOBO biomes in Southeast Asia show a more homogeneous performance profile than South America, with most biomes achieving LOBO ROC-AUC above 0.7, indicating that pan-regional ecological gradients (elevation, forest cover, accessibility) are informative across all biome types. The exception is Mangroves, where designation follows national coastal-zone legislation that produces distinct spatial patterns not well captured by upland-oriented predictors.

Full metrics tables for Southeast Asia are provided in Appendix B.

5.5 Cross-Continental Transfer

To test whether learned patterns are universal or continent-specific, the South America production model is applied to the US test set without retraining, and vice versa, using the common 73-feature set. The results are unambiguous: transfer performance is near-random across all directions. The SA model applied to US test data achieves ROC-AUC = 0.512; the US model applied to SA test data achieves ROC-AUC = 0.487 (effectively the performance of a random classifier). PR-AUC is below 0.01 in all four cross-continental directions.

This finding is substantively informative. It implies that the drivers of PA designation are not universal geographic constants but are instead shaped by continent-level governance structures, land-tenure systems, and conservation histories that produce fundamentally different spatial patterns. What predicts designation in the Amazon does not predict designation in the Rocky Mountains. Even though the feature set is identical (elevation, climate, biodiversity, infrastructure distances), the mapping from features to designation probability is entirely different across continents.

A corollary is that reliable transition-risk assessments require region-specific models: a single global model would fail to capture the institutional specificity of each governance context.

5.6 Forward Projection: Consistency Check

Before presenting forward projections, we verify that the deployment model — trained on 2001–2019 — produces spatially consistent risk scores relative to the evaluation model trained on 2000–2013. This is a prerequisite for trusting the forward results: if the two models assigned fundamentally different risk rankings to the same pixels, the forward deployment model would be extrapolating an unstable pattern.

The per-pixel Pearson correlation between evaluation-model scores and deployment-model scores across South America is $r = 0.34$ ($n = 46.8$ million pixels). This moderate correlation indicates that the rank order of risky pixels is broadly preserved across training windows: pixels that the 2000–2013 model identified as high-risk tend also to be high-risk in the 2001–2019 model. The correlation is not 1.0 because covariate values have changed between 2013 and 2019 (deforestation patterns, PA network expansion, population growth), and because the deployment model has learned from additional years of designation history. This is expected and acceptable: the two models represent different training windows, so some divergence in scores reflects genuine covariate updating rather than model instability. The moderate $r = 0.34$ is sufficient to rule out the possibility that the forward model is extrapolating a noise pattern. A value below 1.0 is expected: covariate values evolve between the two training windows as deforestation, PA network expansion, and population dynamics unfold, so some divergence in scores reflects genuine landscape change rather than model instability.

5.7 Forward Projection (2025–2030)

5.7.1 On-Track Assessment

Table 4 summarises the current protection level, the business-as-usual (BAU) trajectory, and the implied gap to the 30% target for each region.

Table 4: On-track assessment for the 30×30 target. BAU trajectory extrapolates the 2019–2024 designation rate forward.

Region	Current coverage	Gap to 30%	BAU designation rate	Implied shortfall
South America	25.6%	4.4 pp	0.065 pp/yr	∼67 years late
United States	7.7%	22.3 pp	<0.13 pp/yr	>170 years late
Southeast Asia	14.0%	16.0 pp	<0.09 pp/yr	>170 years late

Under the business-as-usual scenario, no region is on track. South America faces the smallest gap, having already reached 25.6% protection, but even at its recent designation pace it would not reach 30% until approximately 2093. The United States and Southeast Asia face far larger deficits relative to their target: both would require more than 170 additional years at current rates.

5.7.2 Scenario Maps and Hotspot Clusters

The forward model identifies spatially explicit priority areas under each scenario. Under the 30×30 scenario, the South America model designates the top-probability pixels until the cumulative area reaches 30% of terrestrial coverage. Spatial clustering (DBSCAN) on these highest-risk pixels identifies 40 distinct clusters in South America (Figure 6), 39 in the United States, and 9 in Southeast Asia. The relatively small cluster count in Southeast Asia reflects the concentration of high-risk pixels in a few large forested upland areas.

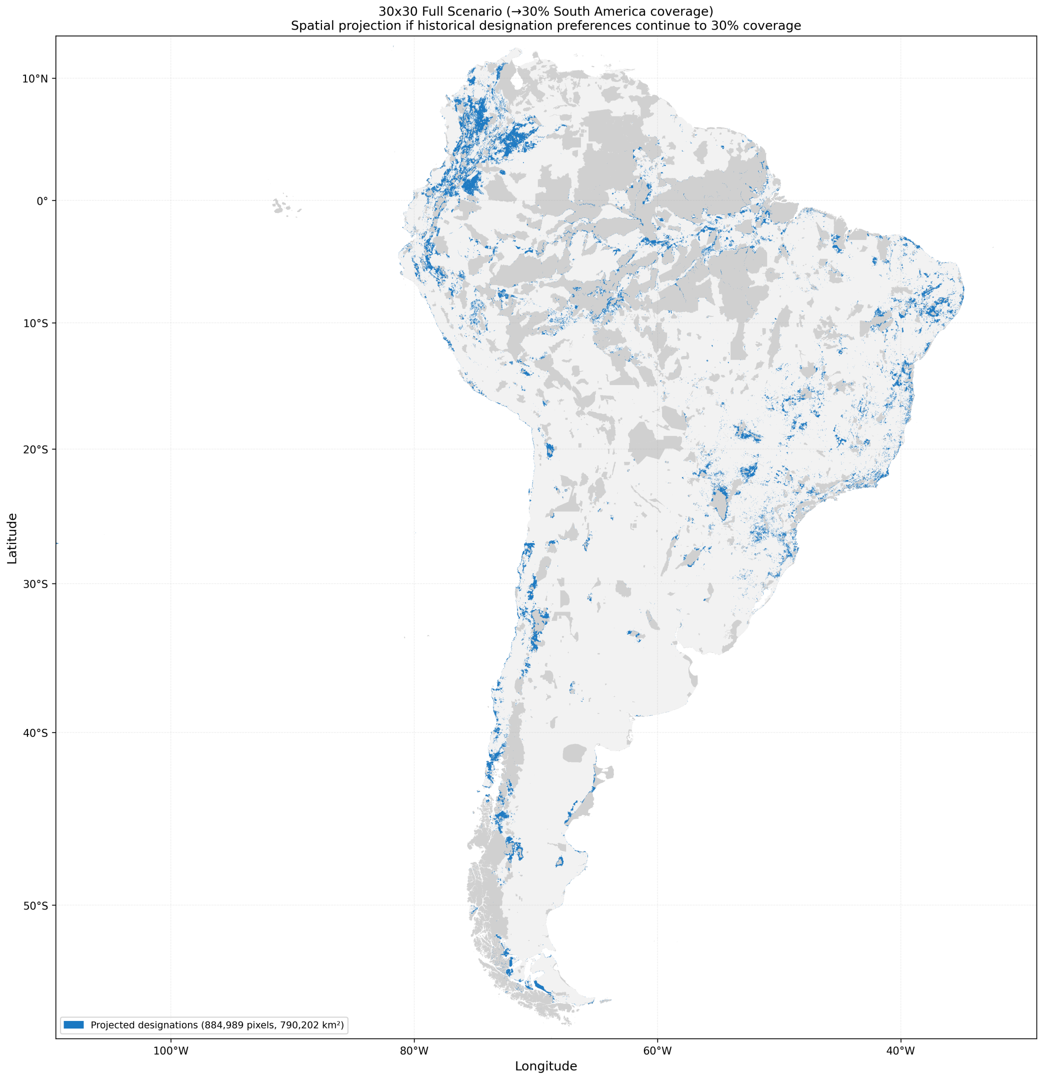

Figure 6: Forward projection for South America under the 30×30 scenario. Coloured clusters indicate spatially contiguous groups of high-probability pixels that would need to be designated to reach 30% terrestrial protection by 2030. Grey background shows the current protected-area network.

In South America, the highest-risk clusters concentrate in the Southern Amazon (Rondônia, Mato Grosso, Pará), the Colombian Amazon, the Cerrado–Amazon transition zone, and the Andes–Amazon piedmont — precisely the areas that have seen the most active designation over 2015–2024. This convergence between predicted forward risk and recent historical patterns provides an additional out-of-sample validation of the model.

The United States clusters concentrate in the Desert Southwest (Arizona, Nevada, Utah), the Pacific Northwest, and the Southeast — areas with large areas of BLM and National

Forest land that historically receive new wilderness and monument designations.

Southeast Asia clusters are concentrated in northern and central Borneo, Sumatra's Bukit Barisan range, and the Philippines' mountain biomes.

5.7.3 Economic Exposure in High-Risk Zones

As a proxy for transition-risk relevance to economic actors, we quantify the area of agricultural land falling within the top-1% risk zone, using MODIS land-cover classifications. In South America, 582 km^2 of agricultural land falls within the highest-risk 1% of unprotected pixels ($\approx$ 156,000 km^2 total). The low absolute share (0.4% of the top-1% zone) reflects the dominance of remote forested and shrubland pixels in the high-risk zone, consistent with the opportunity-cost mechanism documented in Section 5.2.4: land with high agricultural value is actively suppressed from high-risk zones by the model. In the United States, 5,234 km^2 of agricultural land falls within the top-1% risk zone — substantially higher, reflecting both the larger absolute zone area and the greater proportion of mixed agricultural-public land in the US West. In Southeast Asia, the figure is 37 km^2, again reflecting the concentration of high-risk pixels in remote forested uplands far from agricultural zones. These estimates provide a coarse first approximation of the agricultural assets most immediately exposed to designation risk; more granular asset-level analysis would require parcel-level data beyond the scope of this thesis.

6 Discussion

6.1 Driver Hierarchy and the Political Economy of Designation

The SHAP analysis reveals a driver hierarchy in which biodiversity value and governance quality are the primary determinants of designation, with economic opportunity cost entering as a secondary but substantive constraint. GSN conservation-priority indicators rank above all other variable groups; governance indicators (WGI Rule of Law, V-Dem Liberal Democracy Index) rank second, ahead of infrastructure-based proxies for economic activity. This is not the purely cost-driven "least-resistance" story that early empirical work suggested (Joppa and Pfaff, 2009; Venter et al., 2018): rather, institutional capacity to act on conservation commitments appears to be a binding constraint of comparable or greater importance than the economic cost of designation.

The implication for the 30×30 target is therefore twofold. First, in countries with weak institutions, even ecologically important areas may remain unprotected not because they are economically valuable but because the state lacks the administrative capacity to gazette and enforce new PAs. Second, in countries with strong governance, the remaining unprotected biodiversity-priority land tends to be economically valuable — requiring the compensation mechanisms (direct payments, conservation easements, carbon market incentives) that are central to the political economy of ambitious designation. If future designations follow the same governance-and-ecology logic as historical ones, the additional 12 percentage points of coverage will concentrate in countries with both strong institutions and available low-cost land, leaving biodiversity-rich areas in governance-constrained contexts systematically underrepresented (Venter et al., 2018).

The SA PR-AUC of 0.645 — against a random baseline of 0.008 — is consistent with the level of discrimination achieved by Mouillot et al. (2024) using a comparable socio-environmental niche approach, providing cross-study validation that geographic and institutional covariates contain sufficient signal to predict PA designation at continental scale.

6.2 The Representation Gap

The LOBO analysis reveals a predictable pattern of biome underrepresentation under historical dynamics. Biomes with consistently high LOBO performance — arid deserts, certain tropical forests — are those where designation follows regular geographic gradients and where the economic opportunity cost of protection is low. Biomes with poor LOBO performance — temperate conifer forests in the US, montane grasslands in South America — exhibit idiosyncratic designation patterns driven by institutional and political factors that the model cannot capture from generic geospatial covariates alone.

This implies a representation gap: even if 30% terrestrial coverage is achieved by 2030, the distribution of new PAs across biomes is likely to remain biased toward ecologically "easy" land. Closing this gap would require active policy intervention — conservation finance instruments, direct payments to landowners, or binding biome- specific designation targets — that shift designations toward high-opportunity-cost areas.

6.3 Cross-Continental Consistency and Limits of Generalisability

The near-random cross-continental transfer performance (SA→USA ROC-AUC 0.512; USA→SA ROC-AUC 0.487) establishes a clear boundary for the model's generalisability. Protected-area designation patterns are not universal laws derivable from physical geography alone; they are products of specific national legal frameworks, political histories, and institutional capacities. Even a feature set that includes both remotely-sensed covariates and country-level governance indicators cannot capture the specific congressional acts that create a national monument in Utah or the presidential decrees that establish Amazonian reserves — because the *mapping* from general institutional quality to specific spatial designation patterns differs fundamentally across governance regimes.

This finding has two practical implications. For risk assessment, it confirms that region-specific models are required: a single global risk score would obscure continent-level variation in the drivers of designation. For research, it suggests that the most valuable next-generation features are spatially granular institutional ones: parcel-level land-tenure status (public vs. private ownership), which country-level governance indicators cannot proxy, carbon-market eligibility at the project level, and NGO funding flows that enable or constrain specific designation processes.

6.4 Calibrated Probabilities as Transition-Risk Scores

The core value proposition of this framework for financial risk applications depends on calibration. A raw ranking of pixels by predicted score is useful for conservation prioritisation, but transition-risk quantification requires statements of the form "this parcel has a 3% probability of being designated within five years" — an interval-scale probability, not an ordinal rank.

After Platt scaling, the ECE of 0.0081 in South America confirms that the calibrated scores have this property: on average, the predicted probability deviates from the observed transition rate by less than one percentage point across the score distribution. A lender extending a five-year agricultural loan secured on land in the top-1% risk zone could use the calibrated $\hat{p}$ directly to adjust the expected return, price a covenant trigger, or flag the exposure for additional due diligence — applications analogous to the use of credit default probabilities in fixed-income risk management.

The model produces a conditional forecast, not an unconditional one: probabilities are calibrated to historical designation patterns and assume that the policy environment does not change structurally. If the Kunming-Montreal commitments trigger a sustained acceleration in designation pace, calibrated probabilities would underestimate true risk. In this sense, the BAU scenario is a lower bound on transition risk rather than a point forecast.

6.5 Forward Projection and Policy Implications

The on-track assessment in Table 4 delivers a clear message: no region studied here will reach the 30×30 target on time under business-as-usual trajectories. South America, despite having the smallest percentage-point gap, would arrive approximately 67 years late; the United States and Southeast Asia face deficits of more than 170 years at current rates.

Closing this gap requires not only more designations but also different designations. The scenario maps identify which specific geographic clusters would need to be designated under each pathway. Governments committed to the 30×30 target can use these maps in two complementary ways. First, as a prioritisation tool: the high-probability clusters identify which areas are most consistent with historical designation patterns and thus most politically feasible to protect in the near term. Second, as a gap analysis: comparing the scenario maps to biodiversity priority maps (Key Biodiversity Areas, IUCN Red List habitat maps) reveals which high-priority areas are *not* predicted to be designated under business-as-usual — the so-called representation gap — and therefore require special policy attention.

The distinction between governments that need to *accelerate* designation and those that need to *redirect* it matters. South America needs primarily to accelerate; the US needs to redirect as well as accelerate, because the existing gap is too large to fill by protecting only the historically favoured desert and shrubland landscapes.

6.6 Limitations

Stationarity. The model extrapolates historical designation patterns forward. Structural breaks — a change in government, a new carbon-market mechanism, the Kunming-Montreal momentum itself — are not modelled. Calibrated probabilities should be interpreted as conditional forecasts: "what would the five-year designation probability be, given that the historical pattern continues?" They are not unconditional predictions of what will happen.

Missing predictors. The current feature set includes country-level governance indicators (WGI, V-Dem, DPI) and indigenous territory boundaries (LandMark), but remains missing key spatially granular institutional variables: parcel-level land-tenure status (public vs. private ownership) and carbon-market project eligibility (REDD+ credits). Country-level governance quality imperfectly proxies the parcel-level tenure constraints that drive many designation decisions, particularly in mixed federal-private landscapes such as the US West. NGO funding flows — which directly enable specific designation processes — are also unobserved. Incorporating parcel-level tenure data, where available, would be the highest-priority extension.

WDPA dating uncertainty. Gazette year entries in the WDPA may lag the formal legal designation by one to three years in some jurisdictions. The five-year lookahead window reduces sensitivity to small date errors, but systematic lags in data entry could introduce a small bias toward later observation of transitions.

OECMs and target-variable heterogeneity. The WDPA's increasing incorporation of Other Effective Area-Based Conservation Measures (OECMs) — which vary widely in protection strength, from strictly managed nature reserves to loosely governed community areas — may introduce heterogeneity in the target variable not captured by the binary protected/unprotected classification. If OECMs are systematically different in their geographic distribution from traditional PAs, this could bias the learned designation pattern.

Infrastructure as confounder. Road proximity and night-time lights are strong predictors partly because they proxy accessibility, but also partly because they are correlated with economic development levels that independently influence the political feasibility of protection. The model cannot separately identify these mechanisms, and causal interpretation of the feature importance is therefore not warranted.

Non-causal nature. The framework is explicitly predictive, not causal. It cannot evaluate counterfactual policy interventions ("what would designation rates be if carbon credits were tripled?") or estimate the welfare effects of designation. These questions require structural econometric models, which are beyond the scope of this thesis.

7 Conclusion

This thesis has developed and evaluated a machine-learning framework for predicting the spatial expansion of protected areas under the 30×30 biodiversity target, applied to three major world regions. The primary model for South America achieves a ROC-AUC of 0.897 and captures 90.4% of all test-period PA establishments within the top 1% of predicted risk — performance sufficient to make the model a practically useful tool for spatial transition-risk screening. Calibrated predicted probabilities serve as interval-scale transition-risk scores that can be used directly in financial risk quantification, portfolio screening, and land-use planning.

Three substantive findings emerge from the analysis. First, the driver hierarchy reveals that conservation priority (GSN biodiversity indicators) and governance quality (WGI Rule of Law, V-Dem Liberal Democracy Index) are the primary determinants of where PAs are designated — not merely economic opportunity cost. This implies that the path to closing the 30×30 gap runs through strengthening institutions and providing compensation mechanisms in biodiversity-rich, economically contested areas, not only through designating remote land. Second, forward projections show that all three studied regions are severely off-track — South America by approximately 67 years, the United States and Southeast Asia by more than 170 years at current designation rates. Third, cross-continental transfer performance is near random, confirming that designation patterns are products of specific governance histories and institutional contexts, not universal ecological gradients, and that region-specific models are required for credible risk assessment.

For governments, the most actionable output is the scenario maps: they identify specific geographic clusters that would need to be designated under BAU, moderate, and full 30×30 scenarios, and can be compared against biodiversity priority maps to distinguish between acceleration-only and acceleration-plus-redirection policy needs. For investors and lenders, the calibrated probability surface provides a readily deployable transition-risk score for any land parcel in the three regions studied. For researchers, the governance-as-primary-driver finding motivates models that add parcel-level land tenure (public vs. private), carbon-market project eligibility, and NGO funding flows — the spatially granular institutional predictors that country-level governance scores cannot proxy.

The most important extension is the inclusion of parcel-level land-tenure data. Even though the current model includes country-level governance indicators, parcel-level public versus private ownership is qualitatively different in its direct effect on political feasibility, and its absence is the most likely proximate cause of cross-continental transfer failure. A second valuable extension is to update the calibration annually as new WDPA data become available, maintaining probability scores that reflect the current pace of policy implementation rather than historical averages.

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Appendix

A Full LOBO Tables

Tables 5, 6, and 7 report per-biome LOBO ROC-AUC for all three regions (LightGBM models).

Table 5: South America: Per-biome LOBO ROC-AUC (LightGBM). Mean = 0.808.

Biome	N positives	LOBO ROC-AUC
Deserts & Xeric Shrublands	109,678	0.995
Flooded Grasslands & Savannas	1,425	0.801
Mangroves	1,457	0.980
Mediterranean Forests, Woodlands & Scrub	442	0.701
Montane Grasslands & Shrublands	5,184	0.797
Temperate Broadleaf & Mixed Forests	889	0.491
Tropical & Subtropical Dry Broadleaf Forests	10,328	0.847
Tropical & Subtropical Grasslands, Savannas	4,184	0.842
Tropical & Subtropical Moist Broadleaf	62,359	0.814
Mean		**0.808**

Table 6: United States: Per-biome LOBO ROC-AUC (LightGBM). Mean = 0.510 (evaluated biomes only). Four biomes had zero test-period designations and could not be evaluated (N/A).

Biome	N positives	LOBO ROC-AUC
Deserts & Xeric Shrublands	12,625	0.387
Mediterranean Forests, Woodlands & Scrub	216	0.546
Temperate Broadleaf & Mixed Forests	564	0.746
Temperate Conifer Forests	737	0.334
Temperate Grasslands, Savannas & Shrublands	363	0.536
Flooded Grasslands; Mangroves; Trop. Conifers; Trop. Grasslands	0	N/A
Mean (evaluated)		**0.510**

Table 7: Southeast Asia: Per-biome LOBO ROC-AUC (LightGBM). Mean = 0.883 (evaluated biomes only). Three biomes had zero test-period designations.

Biome	N positives	LOBO ROC-AUC
Mangroves	46	0.957
Temperate Conifer Forests	4,284	0.822
Tropical & Subtropical Coniferous Forests	21	0.991
Tropical & Subtropical Dry Broadleaf Forests	17,702	0.818
Tropical & Subtropical Moist Broadleaf Forests	51,475	0.826
Montane Grasslands; Temperate Broadleaf; Trop. Grasslands	0	N/A
Mean (evaluated)		**0.883**

B USA and SE Asia Full Evaluation Tables

Table 8: United States: LightGBM predictive performance on the temporal test set (2017–2019). Baseline positive rate: ≈ 0.040%.

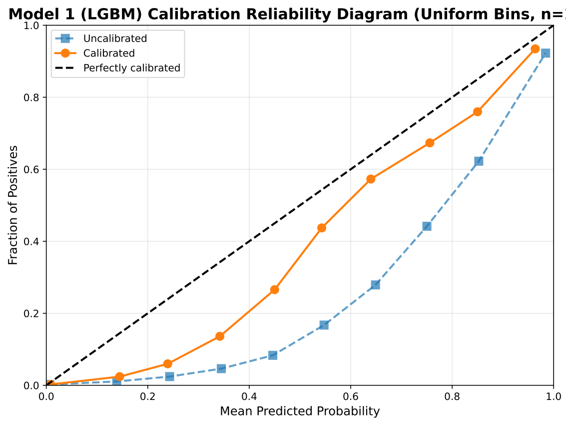
Metric	LightGBM
ROC-AUC	0.944
PR-AUC	0.082
Precision @ top 1%	0.032
Lift @ top 1%	80.6×
Precision @ top 5%	0.007
Lift @ top 5%	17.9×
Brier score	0.000400
ECE (post-calibration)	0.000310

Table 9: Southeast Asia: LightGBM predictive performance on the temporal test set (2017–2019). Baseline positive rate: ≈ 0.596%.

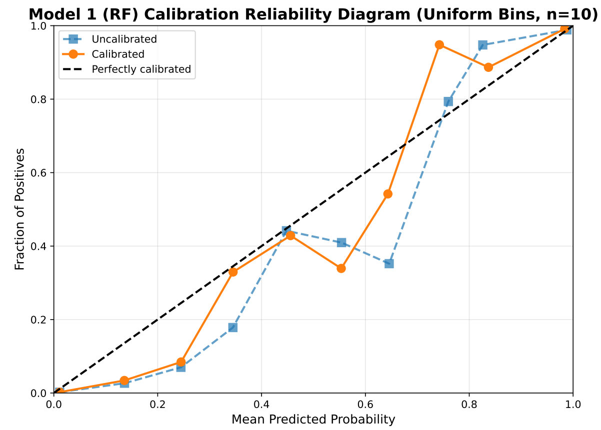
Metric	LightGBM
ROC-AUC	0.994
PR-AUC	0.919
Precision @ top 1%	0.561
Lift @ top 1%	94.2×
Precision @ top 5%	0.115
Lift @ top 5%	19.3×
Brier score	0.00144
ECE (post-calibration)	0.00863

C Reliability Diagrams

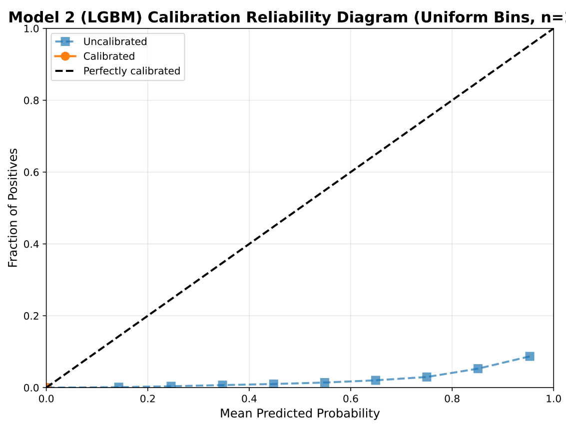
Reliability diagrams show the fraction of positive outcomes (observed transition rate) binned by predicted probability, both before and after Platt scaling (Platt, 1999). Post-calibration probabilities are well-aligned with observed rates across all regions, with ECE values below 0.011 for all LightGBM models.



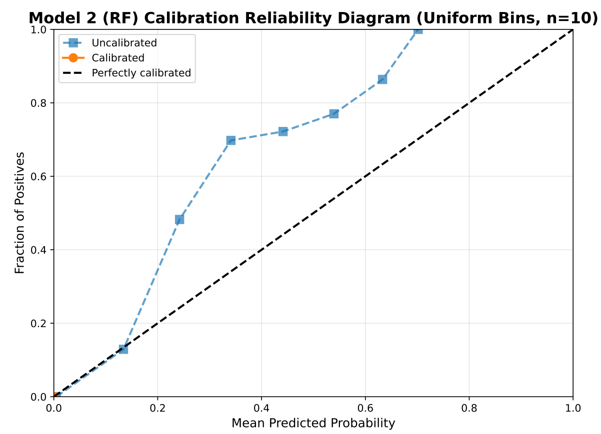
(a) South America — LightGBM



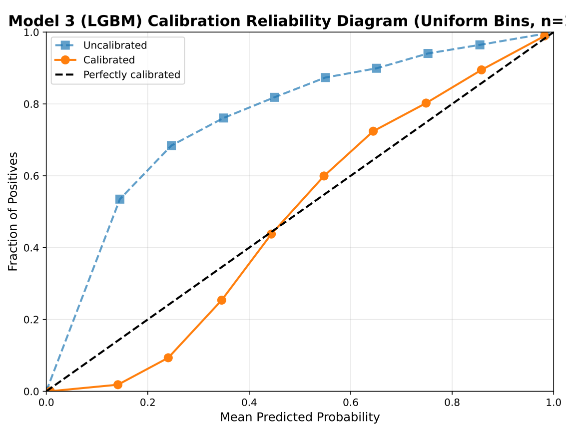
(b) South America — Random Forest



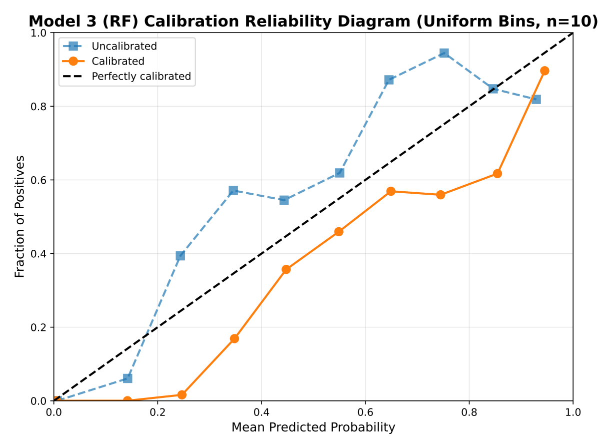
(c) United States — LightGBM



(d) United States — Random Forest



(e) Southeast Asia — LightGBM



(f) Southeast Asia — Random Forest

Figure 7: Reliability diagrams for all six model variants (before and after Platt scaling). Bars show the mean predicted probability within each bin; the diagonal represents perfect calibration.

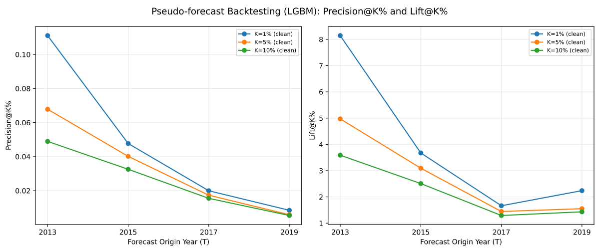
D Cross-Continental Transfer Detail

The cross-continental transfer (SA→USA and USA→SA) is implemented using the 73-feature set common to both regions. The SA production model is serialised after training on 2001–2019 South America data and scored on the US test panel (2017–2019) without any retraining. The same procedure is repeated in the reverse direction.

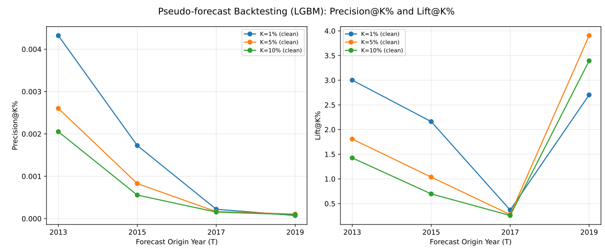
Transfer ROC-AUC values: SA→USA = 0.512; USA→SA = 0.487. PR-AUC < 0.01 in both directions. The LightGBM result is sufficient for the cross-continental inference: because the LGBM and RF models make qualitatively similar ranking decisions within each region (Jaccard overlap at top-1%: SA = 0.262, USA = 0.121), near-random transfer performance in either model establishes the same conclusion. The RF cross-continental transfer is therefore omitted; the LGBM transfer result stands as the primary finding.

E Backtesting Results

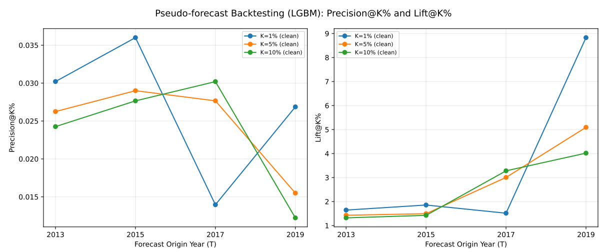
Walk-forward backtests at origins $T \in \{2009, 2011\}$ verify that deployment-style inference generalises to out-of-sample designation events. Each origin trains on 2001–$(T-1)$, scores unprotected pixels in year T, and evaluates against realised designations in years $T+1$ through $T+5$. SA LightGBM backtest precision@1% is stable across origins; USA precision is lower and more variable, consistent with the test-set PR-AUC.



(a) South America — LightGBM



(b) United States — LightGBM



(c) Southeast Asia — LightGBM

Figure 8: Backtesting precision@1% over time for walk-forward forecasts at origins $T \in \{2009, 2011\}$. Each line shows the precision of the top-1% predicted pixels evaluated against realised 5-year designation outcomes.

F Feature Dictionary

Table 10 lists the full feature set used in the South America, USA, and Southeast Asia models. All three regions share the same 73-feature schema, derived from the predictor groups in Table 1 via multi-scale spatial smoothing (4×4, 16×16, 64×64 km), Euclidean distance transforms, and temporal lags.

Table 10: Feature dictionary (selected entries). Full schema has 73 features after spatial smoothing. *Static* = time-invariant; *Dynamic* = updated annually.

Feature name	**Type**	**Description**
`dist_wdpa`	Dynamic	Euclidean distance to nearest PA boundary (km), lagged 1 yr
`elevation_b1`	Static	Elevation (m, CGIAR SRTM V4)
`slope_b1`	Static	Slope (degrees, derived from SRTM)
`NDVI_b1`	Dynamic	Annual median NDVI (MODIS MOD13Q1)
`NDVI_b1_smooth4`	Dynamic	NDVI averaged over 4×4 km window
`NDVI_b1_smooth16`	Dynamic	NDVI averaged over 16×16 km window

Feature name	Type	Description
NDVI_b1_smooth64	Dynamic	NDVI averaged over 64×64 km window
deforestation_b1	Dynamic	Annual forest-loss indicator (Hansen)
deforestation_b1_smooth64	Dynamic	Forest loss averaged over 64×64 km window
wildfire_b1	Dynamic	Annual burned-area indicator (MODIS MCD64A1)
GPW_b1	Dynamic	Population density (GPW v4, forward-filled)
GPW_b1_smooth16	Dynamic	Population density at 16×16 km
GPW_b1_smooth64	Dynamic	Population density at 64×64 km
HNTL_b1	Dynamic	Night-time lights (HNTL, DMSP/VIIRS harmonised)
HNTL_b1_smooth64	Dynamic	Night-time lights at 64×64 km
dist_road	Static	Distance to nearest major road (km, GRIP)
dist_powerplant	Static	Distance to nearest power plant (km, WRI)
dist_oil_gas	Static	Distance to nearest oil/gas infrastructure (km, Omara et al.)
dist_indigenous	Static	Distance to nearest indigenous/community land (km, LandMark)
landcover_b1	Dynamic	Annual dominant land-cover class (MODIS MCD12Q1)
economic_value_lag1	Dynamic	Agricultural land value (USD/ha, FAO FAOSTAT), lagged 1 yr
policy_b2	Dynamic	V-Dem Liberal Democracy Index (country level)
policy_b3	Dynamic	WGI Government Effectiveness (country level)
policy_b4	Dynamic	WGI Rule of Law (country level)
GSN_b2	Static	GSN high-biodiversity area indicator
GSN_b2_smooth16	Static	GSN biodiversity averaged over 16×16 km
GSN_b2_smooth64	Static	GSN biodiversity averaged over 64×64 km

Feature name	Type	Description
`WorldClim_b7`	Static	WorldClim V1 annual temperature range
`WorldClim_b14`	Static	WorldClim V1 precipitation of driest month
`WorldClim_b17`	Static	WorldClim V1 precipitation seasonality
...	...	Additional WorldClim, GSN, smoothed, and policy variants (73 total)

G Colombia Sub-Region Results

A fully separate pipeline was implemented for Colombia as a computational testbed during early model development. On the Colombia test set (2017–2019), the LightGBM model achieves ROC-AUC 0.946 and PR-AUC 0.470; the Random Forest achieves ROC-AUC 0.931 and PR-AUC 0.435. These results confirm that both model classes perform consistently at smaller geographic scales and that the continental South America model is not simply benefiting from its larger sample size.

The Colombia pipeline is not on the critical path of the main analysis and is included here for completeness.